

Summary of Master's Thesis

Cartographic Modeling of Cadastral Resurvey Effort and Cost: A Case Study in Cape Town, South Africa

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Background and Research Problem

Cadastral surveys accumulated through different surveying eras reflect the standards, measurement practices, coordinate systems and technologies of their time, resulting in heterogeneous records with varying levels of accuracy and characteristics. Consequently, the effort required to resurvey land parcels can vary substantially. This thesis develops a spatial approach to account for these differences when estimating cadastral resurvey cost and effort, using Cape Town, South Africa, as a case study.

The statutory tariff provides a baseline for cadastral survey costs and rules for additional charges associated with factors such as parcel area, number of beacons, terrain and vegetation. However, several of these factors are expressed through qualitative categories, making consistent assessment across a large cadastral dataset difficult and requiring substantial manual investigation. The thesis therefore translates these considerations into quantitative spatial indicators and a GIS-based decision-support model, enabling relative resurvey effort to be estimated and visualised consistently across the study area.

Data and Methodology

The study focuses on the urban cadastral fabric of the City of Cape Town Municipality, comprising approximately 800,000 land parcels. The primary cadastral dataset was obtained from the Office of the Surveyor-General, Western Cape, and linked through the Land Parcel Identifier (LPI) to Surveyor-General (SG) diagram and Survey Record (SR) identifiers. The years associated with these records were used as temporal indicators of cadastral survey activity.

The cadastral data were integrated with a 5 m LiDAR-derived Digital Elevation Model (DEM), tree-canopy data, a municipal road network and Town Survey Marks (TSMs), which provide survey control points. These datasets were obtained from the City of Cape Town and its Open Data Portal. The DEM was used to derive slope, canopy data to quantify vegetation coverage, and road-network distances to TSMs to measure access to survey control. The datasets were integrated in PostgreSQL/PostGIS.

For each parcel, measurable indicators were derived for geometric complexity, terrain, vegetation, survey age and access to survey control. These were incorporated into a cost model based on the 2003 Gazette tariff: parcel area determined the base cost, while adjustments represented boundary complexity, topography-vegetation conditions, survey age and TSM proximity. Parcels were further aggregated into approximately 36,000 cadastral blocks for multi-scale analysis.

A supporting Neo4j graph represented parcel-beacon relationships, including shared beacons between parcels, together with blocks, TSMs and allotments. The two nearest TSMs were identified for each parcel using road-network distance to represent survey-control access, and the resulting connections supported exploration of cadastral connectivity independently of the cost calculation.

Results and Contribution

The model was applied across the Cape Town cadastral dataset to produce parcel-level estimates of relative resurvey effort, expressed as unitised cost, defined as the estimated parcel cost relative to the base cost of the simplest parcel before adjustments. Most parcels were relatively simple in geometric and environmental terms. The resulting model outputs confirmed the intended hierarchy of factor influence. Geometric complexity was dominant, with parcel sides showing a strong correlation with unitised cost ($r = 0.92$), while the other factors showed weaker associations.

The resulting parcel-level cost estimates showed clear spatial patterns across Cape Town. High-cost areas such as Clifton, Chapman's Peak and Bishopscourt were associated with steep terrain, dense vegetation and irregular cadastral geometry, while planned townships such as Mitchells Plain and Delft in the Cape Flats showed lower costs, reflecting regular parcel layouts and minimal topographic or vegetation interference. At allotment level, Clifton had a mean unitised cost of 2.05 versus 0.95 for Mitchells Plain.

Aggregation also produced a multi-scale representation of resurvey effort, with broader neighbourhood and township patterns emerging at block and allotment levels and local extremes becoming less pronounced.

The thesis contributes a spatially explicit framework for estimating cadastral resurvey effort by translating cost-related surveying considerations into measurable geospatial variables. Its main contribution is integrating cadastral geometry and history with terrain, vegetation and survey-control access within a GIS-based model, enabling consistent assessment across a large cadastral dataset.

A further contribution is the adaptation of the statutory cadastral survey tariff into a spatial decision-support model. Rather than treating the tariff solely as a tabular costing mechanism, the research operationalises its relevant components together with additional spatial factors and applies them systematically at parcel level. The resulting PostgreSQL/PostGIS database supports analysis across parcel, block and allotment-township scales, while the associated Neo4j graph represents relationships among cadastral entities and survey control.

Limitations and Future Directions

The model is constrained by the completeness and consistency of the available cadastral and survey records. SG information covered 96.4% of parcels, while SR coverage was substantially lower, limiting the availability of detailed survey-record information for modelling parcel history. The model also relies on the 2003 Gazette tariff as its cost baseline and therefore represents a structured estimate of relative resurvey cost rather than a prediction of current market prices or actual survey expenditure.

Environmental and accessibility factors were represented through spatial proxies, including slope, tree-canopy coverage and road-network distance to TSMs. These provide consistent measures but simplify some conditions encountered during actual resurvey. The model was developed for Cape Town, and application to other cadastral systems would depend on comparable cadastral, survey, environmental and tariff information.

The framework moves assessment from isolated parcel-by-parcel investigation towards a structured, multi-scale approach supporting prioritisation and planning at parcel, block and allotment-township levels. Its integration within PostgreSQL/PostGIS and a graph representation of cadastral relationships provides a foundation for further interconnected spatial information systems.

The research provides a foundation for further work in geospatial data integration, spatial databases and urban modelling. Its integration of cadastral, environmental, temporal and network datasets demonstrates experience in structured spatial workflows and decision support. These capabilities are transferable to urban-scale geospatial systems and future research in 3D geospatial modelling, digital twins, dynamic spatial databases and integrated urban analysis.